

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Biology — Class 9 | Exam-14 | Total Marks: 43

Syllabus: Chapter 1 — The Science of Biology | Chapter 2 — Biodiversity | Chapter 3 — The Cell | Chapter 4 — Cell Cycle | Chapter 5 — Tissues, Organs & Organ Systems | Chapter 6 — Molecular Biology | Chapter 7 — Metabolism (till 7.6 only)

Every question on this paper is traceable to Seemab's own textbook. Each MCQ below carries the exact Student Learning Outcome (SLO) or section it comes from. No question uses general knowledge, no question comes from another subject's book, and nothing falls outside the studied syllabus.

SECTION A — MCQ Answers (24 × 1 = 24 Marks)

1. The word **Biology** comes from two Greek words *bios* and *logos*, meaning: → **A. life and study** 1
bios = life, *logos* = thought, reasoning and study. Biology is the study of living organisms.
Syllabus reference: Ch1 SLO 1; §1.1.1

2. The branch of biology that deals with the study of animals is: → **B. Zoology** 1
The three major fields are Botany (plants), **Zoology (animals)** and Microbiology (micro-organisms).
Syllabus reference: Ch1 SLO 3

3. The branch of biology dealing with the study of **tissues** is: → **C. Histology** 1
Histology is the study of tissues. Cytology studies cells, Anatomy internal structure, Physiology functions.
Syllabus reference: Ch1 SLO 4

4. The variety of living organisms present in an area is called: → **B. biodiversity** 1
Biodiversity is the variety of living organisms. Classification is the grouping of those organisms.
Syllabus reference: Ch2 SLO 1

5. Living organisms are broadly classified into three domains: Archaea, Bacteria and: → **C. Eukarya** 1
The three domains are **Archaea, Bacteria and Eukarya**. Fungi, Protista and Plantae are kingdoms, not domains.
Syllabus reference: Ch2 SLO 4

6. In the scientific name *Homo sapiens*, the word *Homo* is the: → **B. genus** 1
In binomial nomenclature the **first word is the genus** (capitalised) and the second is the species.
Syllabus reference: Ch2 SLO 7

7. The basic structural and functional unit of life is the: → **B. cell** 1
The **cell** is the basic unit of life. Organelles are parts within a cell; tissues are groups of cells.
Syllabus reference: Ch3 SLO 1

8. The organelle that controls all the activities of a cell is the: → **B. nucleus** 1
The **nucleus** contains the genetic material and controls all cell activities.
Syllabus reference: Ch3 SLO 3

9. Unspecialized cells that are able to develop into different types of cells are called: → **B. stem cells** 1
Stem cells are unspecialized and can become specialized cell types.
Syllabus reference: Ch3 SLO 10

10. Which type of cell is specialised for carrying messages within the body? → **B. neuron** 1
A **neuron** (nerve cell) is specialised to carry messages. Red blood cells transport oxygen.
Syllabus reference: Ch3 SLO 5

11. During which phase of the cell cycle is DNA replicated? → **B. S phase** 1
DNA is copied during the **S (synthesis) phase** of interphase.
Syllabus reference: Ch4 SLO 1

12. The correct order of the stages of mitosis is: → **A. prophase, metaphase, anaphase, telophase** 1

P.M.A.T. — Prophase, Metaphase, Anaphase, Telophase.

Syllabus reference: Ch4 SLO 2

13. Meiosis produces: → **B. four genetically different haploid cells** 1

Meiosis gives **four haploid cells that are genetically different**. Mitosis gives two identical diploid cells.

Syllabus reference: Ch4 SLO 3

14. Which sequence shows the correct increasing order of organisation in the body? → **A. cells → tissues → organs → organ systems** 1

Each level gains new functions: **cells → tissues → organs → organ systems → organism**.

Syllabus reference: Ch5 SLO 1

15. A group of similar cells working together to perform the same function is called: → **B. a tissue** 1

A **tissue** is a group of similar cells with a common function. Several tissues form an organ.

Syllabus reference: Ch5 SLO 2

16. Which one of the following is an **organ** of a plant? → **B. leaf** 1

The **leaf** is a plant organ made of several tissues. Xylem is a tissue, mesophyll a cell type, chloroplast an organelle.

Syllabus reference: Ch5 SLO 8

17. Which biomolecule is the main carrier of hereditary information? → **C. DNA** 1

DNA stores and carries hereditary information from one generation to the next.

Syllabus reference: Ch6 SLO 8

18. The building blocks (monomers) of proteins are: → **B. amino acids** 1

Proteins are made of **amino acids** joined together. Nucleotides build DNA and RNA.

Syllabus reference: Ch6 SLO 3

19. Glucose is an example of a: → **A. monosaccharide** 1

Glucose is a single sugar unit, a **monosaccharide**. Sucrose is a disaccharide, starch a polysaccharide.

Syllabus reference: Ch6 SLO 6

20. In a DNA molecule, adenine always pairs with: → **C. thymine** 1

Complementary base pairing: **A pairs with T**, and G pairs with C. Uracil occurs in RNA, not DNA.

Syllabus reference: Ch6 SLO 7

21. Unlike a DNA molecule, an RNA molecule is: → **B. single stranded** 1

RNA is a **single stranded** macromolecule of nucleotides; DNA is a double helix.

Syllabus reference: Ch6 SLO 9

22. The building up of complex molecules from simpler ones inside the body is called: → **B. anabolism** 1

Anabolism builds up; catabolism breaks down. Together they make up metabolism.

Syllabus reference: Ch7 SLO 1

23. Chemically, enzymes are: → **B. proteins** 1

Enzymes are **proteins** that act as biological catalysts and are not used up in the reaction.

Syllabus reference: Ch7 SLO 2

24. In **competitive inhibition**, the inhibitor binds to the: → **A. active site of the enzyme** 1

A competitive inhibitor resembles the substrate and occupies the **active site**. A non-competitive inhibitor binds elsewhere.

Syllabus reference: Ch7 SLO 5

SECTION B — Short Question Answers (3 × 3 = 9 Marks)

Q.2 (i) Taxonomic ranks and the meaning of species. 3

Taxonomic ranks, largest to smallest: Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species 2

Species: a group of organisms that share common characteristics and can **interbreed to produce fertile offspring. 1**

Syllabus reference: Ch2 SLO 5 (list the taxonomic ranks) and SLO 6 (define species).

Q.2 (ii) Structure of DNA, base pairing and function. 3

Structure: DNA is a **double helix** macromolecule made of **nucleotides**. Two strands coil around each other, held together by bonds between the nitrogenous bases. **1**

Complementary base pairing: adenine (A) pairs with thymine (T), and guanine (G) pairs with cytosine (C). 1

Function: DNA acts as the **carrier of hereditary information**, passing characteristics from parents to offspring. **1**

Syllabus reference: Ch6 SLO 7 (structure of DNA, complementary base pairing) and SLO 8 (DNA as carrier of hereditary information).

Q.2 (iii) Factors influencing enzyme activity. 3

Any **three of: temperature, pH, substrate concentration, enzyme concentration. 2**

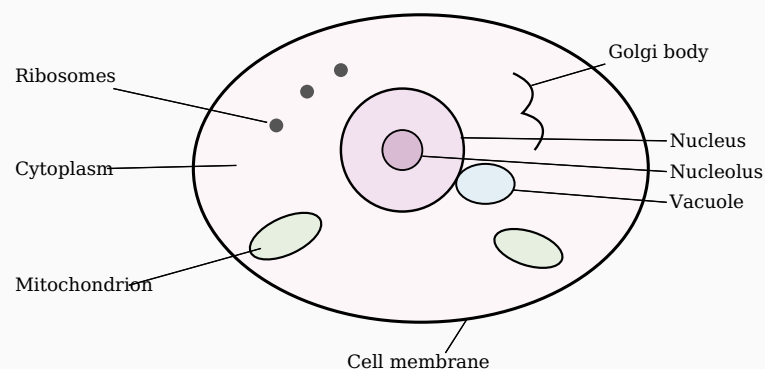
Explanation of one (temperature): as temperature rises, enzyme activity increases because molecules collide more often, until the **optimum temperature** is reached. Beyond that the enzyme **denatures**: its shape, including the active site, is destroyed, so the substrate no longer fits and activity falls sharply. **1**
(An answer based on pH is equally acceptable: each enzyme works best at an optimum pH and denatures away from it.)

Syllabus reference: Ch7 SLO 4 (assess the factors which could influence enzyme activity). Within the studied portion, Ch7 up to 7.6.

SECTION C — Long Question Answers (2 × 5 = 10 Marks)

Q.3 (i) Animal cell diagram and comparison with a plant cell. 5

(a) Well-labelled animal cell 3 — 1 mark for the outline and shape, 2 marks for six correct labels.



(b) Two differences between an animal cell and a plant cell 2

Animal cell	Plant cell
Has only a cell membrane , no cell wall	Has a rigid cell wall outside the membrane
No chloroplasts	Has chloroplasts
Vacuoles are small or absent	Has one large central vacuole

Syllabus reference: Ch3 SLO 2 ("Compare with diagram the structure of animal and plant cell") and SLO 3 (subcellular organelles and their roles).

Q.3 (ii) Stages of mitosis, and mitosis compared with meiosis. **5**

(a) The four stages in order: Prophase → Metaphase → Anaphase → Telophase **1**

Any **two** described **2**:

- **Prophase:** chromosomes condense and become visible, the nuclear membrane starts to break down and the spindle begins to form.
- **Metaphase:** chromosomes line up along the **equator** (centre) of the cell, attached to spindle fibres.
- **Anaphase:** sister chromatids are pulled apart to **opposite poles** of the cell.
- **Telophase:** chromosomes reach the poles, uncoil, and two new nuclear membranes form.

(b) Two differences between mitosis and meiosis **2**

Mitosis	Meiosis
Produces two daughter cells	Produces four daughter cells
Daughter cells are diploid and genetically identical to the parent	Daughter cells are haploid and genetically different
Occurs in body (somatic) cells for growth and repair	Occurs in reproductive cells to form gametes

Syllabus reference: Ch4 SLO 2 (stages of mitosis by sketch and diagram) and SLO 3 (compare mitosis and meiosis).

— End of Answer Key —